

QUESTION 1 (9 Marks)

(a) If $f(x) = x \ln(e^{\sqrt{x}})$, find $f'(x)$ (1)

(b) Find $f(x)$, given that $f''(x) = \frac{3}{\sqrt{x}}$, $f'(4) = 7$ and $f(4) = 20$ (3)

(c) Determine (2)

$$\lim_{x \rightarrow 0} \frac{6x + 6x \cos 6x}{\sin 6x \cos 6x}$$

(d) Use Simpson's rule with 3 function values to estimate (3)

$$\int_{\pi/2}^{\pi} x^2 \sin^2 x \, dx$$

Express your answer to three decimal places.

QUESTION 2 (9 Marks)

(a) Evaluate $\cos^{-1}\left(-\frac{\sqrt{3}}{2}\right) + \tan^{-1}\left(-\frac{1}{\sqrt{3}}\right)$ (2)

(b) (i) Show that the equation of the tangent, ℓ , to $y = e^{3x}$ at $x = 1$ is (2)

$$y = e^3(3x - 2)$$

(ii) If ℓ cuts the x -axis at P and the y -axis at Q , find the coordinates of P and Q . (2)

(iii) Show that the area bound by the y -axis, the curve $y = e^{3x}$ and ℓ is $\frac{1}{6}(5e^3 - 2)$ square units. (3)

QUESTION 3 (9 Marks)

(a) Find $\int \frac{x^2-5}{x^2+3} dx$ (2)

(b) A function is defined by $f(x) = \frac{3x}{x^2+1}$

(i) Find any turning points and hence their nature. (3)

(ii) Hence graph $y = \frac{3x}{x^2+1}$ clearly showing what happens to y as x grows large. (2)

(iii) Find the exact area of the region bounded by the curve, the y -axis, and the line $y = \frac{3}{2}$ (2)

QUESTION 4 (9 Marks)

(a) Use the substitution $u = 4 - x^2$ to show that: (3)

$$\int_0^1 \frac{x}{\sqrt{4-x^2}} dx = 2 - \sqrt{3}$$

(b) (i) Express $\sqrt{3} \sin 2\theta - \cos 2\theta$ in the form $R \sin(2\theta - \alpha)$, $0 \leq \alpha \leq \frac{\pi}{2}$ (2)

(ii) Hence solve: $\sqrt{3} \sin 2\theta - \cos 2\theta = 1$; $0 \leq \theta \leq \pi$. (2)

(c) For what values of m is the line with equation $mx + y = 3$ a tangent to the circle $x^2 + y^2 = 5$? (2)

QUESTION 5 (9 Marks)

(a) Consider the function $f(x) = \frac{1}{2} \cos^{-1}(3x-1)$

(i) State the domain and range of $f(x)$. (1)

(ii) Hence sketch the graph of $y = f(x)$. (1)

(b) In the diagram below, two circles of equal radius r units are drawn such that their centres O and P are r units apart. The two circles intersect at A and B .

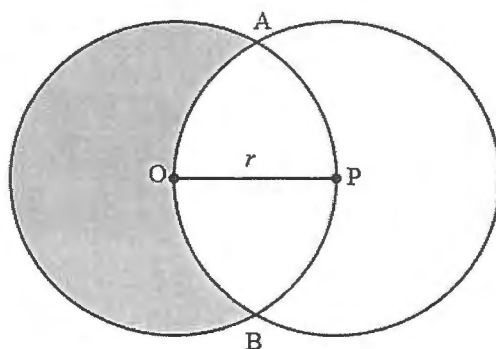


Diagram not to scale

(i) Show that quadrilateral $AOBP$ is a rhombus. (1)

(ii) Hence, or otherwise, find the area of the shaded region in terms of r . (2)

(c) (i) Differentiate with respect to x : $[\tan^{-1}(\frac{x}{3})]^2$ (2)

(ii) Hence find the exact value of (2)

$$\frac{1}{\pi} \int_0^{\sqrt{3}} \frac{\tan^{-1}(\frac{x}{3})}{x^2 + 9} dx$$

QUESTION 6 (9 Marks)

- (a) Given $f(x) = \sin^{-1}(x^2 - 1)$.
- (i) Find $f'(x)$ (1)
- (ii) Hence write down the domain of $f'(x)$ (1)
- (b) Find the exact volume of the solid obtained by rotating the region bounded by the curves $y = 1 - x^2$ and $y = 1 - x$ about the x -axis. (2)
- (c) Prove that $\tan^{-1}\left(\frac{x}{\sqrt{a^2 - x^2}}\right) = \sin^{-1}\left(\frac{x}{a}\right)$, $|x| < a$ (2)
- (d) The cost (C) and revenue (R) of a refrigerator manufacturer can be modelled respectively by the equations (3)

$$C = 87\,000 + 150x \quad \text{and}$$

$$R = x^2 - 80\,500,$$

where x is the number of units produced in 1 week.

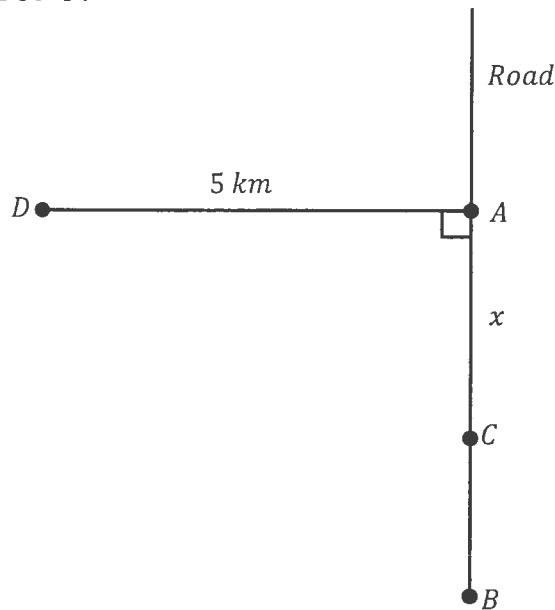
If production in one particular week is 500 units and is increasing at a rate of 300 units per week, find the rate at which the profit is changing.

QUESTION 7 (9 Marks)

- (a) Find the values of x for which the following inequations are simultaneously satisfied:
- $$x + \frac{1}{|x|} < 0 \quad \text{and} \quad x^2 - x - 2 > 0 \quad (3)$$
- (b) The region under the graph $y = \sqrt{x}e^x$ between $x = 1$ and $x = 3$ is rotated about the x -axis. Using the trapezoidal rule with five function values, estimate the volume of the solid formed, to four significant figures. (3)

- (c) A motorist is stranded in the desert 5 kilometres from point A, which is a point on a long straight road nearest to him, as shown in the diagram below. He wishes to get to a point B, on the road, which is 5 kilometres from A. He can travel at 16 km/h on the desert and 39 km/h on the road.

Let the distance from A to C be x kilometres and the time taken to reach his destination be T .



- (i) Show that the time, T , is given by $T = \frac{39\sqrt{25+x^2}-16x+80}{624}$ (1)
- (ii) Hence find the point C at which the motorist must join the road to get to B in the shortest possible time and the time taken to achieve this. (2)

END OF EXAMINATION

V12 MATH EXT1 ASSESSMENT TASK 1
TERM 4, 2012

MATHEMATICS Extension 1 : Question 1		
Suggested Solutions	Marks	Marker's Comments
Q1(a) $f(x) = x \ln(e^{\sqrt{x}})$ $= x\sqrt{x} \ln e$ $= x\sqrt{x}$ $= x^{3/2}$ $\therefore f'(x) = \frac{3}{2}\sqrt{x}$	$f(x) = \ln(e^{\sqrt{x}}) +$ $x \cdot \frac{e^{\sqrt{x}}}{e^{\sqrt{x}}} \times \frac{1}{2\sqrt{x}}$ $= \ln(e^{\sqrt{x}}) + \frac{x}{2\sqrt{x}}$ $= \ln e^{\sqrt{x}} + \frac{\sqrt{x}}{2}$ $=$	1/2 1/2 0M4!!!
(b) $f''(x) = \frac{3}{\sqrt{x}} = 3x^{-\frac{1}{2}}$ $\therefore f'(x) = \int 3x^{-\frac{1}{2}} dx = \frac{3}{\frac{1}{2}} x^{\frac{1}{2}} + C_1 = 6\sqrt{x} + C_1$ but $f'(4) = 7$ so $7 = 6\sqrt{4} + C_1$ $7 = 6 \times 2 + C_1$ $7 = 12 + C_1$ $\therefore C_1 = -5$ $\therefore f(x) = 6\sqrt{x} - 5$ so $f(x) = \int (6x^{\frac{1}{2}} - 5) dx$ $= 4x^{\frac{3}{2}} - 5x + C_2$ but $f(4) = 20$ $\therefore 20 = 4 \times 4^{\frac{3}{2}} - 20 + C_2$ $\Rightarrow C_2 = 8$ $\therefore f(x) = 4x^{\frac{3}{2}} - 5x + 8$	(1/2) 1 1/2 1/2	1 $\frac{6}{\frac{3}{2}} = \frac{6 \times 2}{3} = 4$ $20 = 32 - 20 + C_2$ $C_2 = 8$

MATHEMATICS Extension 1 : Question...

Suggested Solutions

Marks

Marker's Comments

(c) Many Approaches

$$\lim_{x \rightarrow 0} \frac{6x(1+\cos 6x)}{\sin 6x \cos 6x}$$

$$I: \lim_{x \rightarrow 0} \frac{6x}{\sin 6x} \times \lim_{x \rightarrow 0} \frac{(1+\cos 6x)}{\cos 6x}$$

$$= 1 \times \frac{(1+\cos 0)}{\cos 0}$$

$$= 1 \times \frac{(1+1)}{1}$$

$$= 2$$

$$(II): \lim_{x \rightarrow 0} \left(\frac{6x}{\sin 6x \cos 6x} + \frac{6x \cos 6x}{\sin 6x \cos 6x} \right)$$

$$= \lim_{x \rightarrow 0} \left(\frac{12x}{\sin 12x} + \frac{6x}{\sin 6x} \right)$$

$$= 1 + 1$$

$$= 2$$

$$+ \frac{6x \cos 6x}{\sin 6x \cos 6x} \\ + \frac{6x}{\sin 6x} \quad \cos 6x \neq 0 \text{ for } x=0$$

[2]

$$(III): \lim_{x \rightarrow 0} \frac{6x}{\frac{1}{2} \sin 12x} \cdot (2 \cos^2 3x) = \lim_{x \rightarrow 0} \frac{12x}{\sin 12x} \times 2 \cos^2 3x$$

$$= 1 \times 2 \cos^2 0$$

$$= 2$$

$$(d) I = \int_{\frac{\pi}{2}}^{\pi} x^2 \sin^2 x \, dx$$

$$h = \pi - \frac{\pi}{2} = \frac{\pi}{2}$$

$$I \doteq \frac{b-a}{6} \left[f(a) + 4f\left(\frac{a+b}{2}\right) + f(b) \right]$$

$$= \frac{\pi - \frac{\pi}{2}}{6} \left[f\left(\frac{\pi}{2}\right) + 4f\left(\frac{3\pi}{4}\right) + f(\pi) \right]$$

$$= \frac{\pi}{12} \left[\frac{\pi^2}{4} + 4 \times \frac{9\pi^2}{32} + 0 \right] \quad \text{Need the values}$$

$$I \doteq \frac{h}{3} [y_1 + 4y_2 + y_3]$$

$$= \frac{\pi}{3} \left[\frac{\pi^2}{4} + 4 \times \frac{9\pi^2}{32} + 0 \right]$$

$$= \frac{\pi}{12} \left[\frac{\pi^2}{4} + \frac{9\pi^2}{8} \right]$$

$$= \frac{\pi}{12} \times \frac{11\pi^2}{8} \quad \frac{\pi \times 13.5707...}{12}$$

$$= \frac{11\pi^3}{96}$$

$$= 3.5528...$$

display

$$= \frac{\pi}{12} \left[\frac{11\pi^2}{8} \right]$$

$$= \frac{11\pi^3}{96}$$

$$= 3.5528...$$

$$I = 3.553 \text{ (3dp)}$$

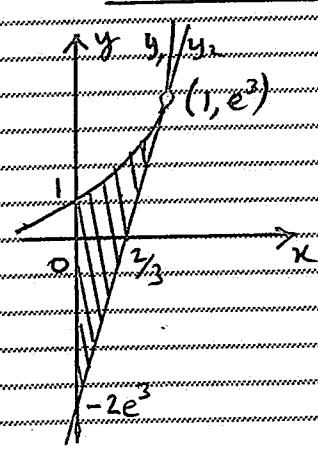
display' ($\frac{1}{2}$)

$\frac{1}{2}$ (1)

[3]

$$\therefore I = 3.553 \text{ (3dp)}$$

Note: NOT 'AREA' question!!

Suggested Solutions	Marks	Marker's Comments
a) $\cos^{-1}\left(\frac{-\sqrt{3}}{2}\right) = \frac{5\pi}{6} \quad (0 \leq \cos^{-1}x \leq \pi)$ $\tan^{-1}\left(\frac{-1}{\sqrt{3}}\right) = -\frac{\pi}{6} \quad (-\frac{\pi}{2} < \tan^{-1}x < \frac{\pi}{2})$ $\therefore \cos^{-1}\left(\frac{-\sqrt{3}}{2}\right) + \tan^{-1}\left(\frac{-1}{\sqrt{3}}\right) = \frac{5\pi}{6} - \frac{\pi}{6} = \frac{2\pi}{3}$	1 1	Too many students looked for something harder.
b) On $y = e^{3x}$, when $x=1$, $y=e^3$ $\frac{dy}{dx} = 3e^{3x} \therefore m = 3e^3$ when $x=1$. $\therefore$ Tangent is $(y - e^3) = 3e^3(x - 1)$ $y = 3e^3x - 3e^3 + e^3$ <u>$y = e^3(3x - 2)$</u>	$\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$	Nearly all students got full marks here
c) At P, $y=0 \therefore 3x-2=0 \therefore P = (\frac{2}{3}, 0)$ $x = \frac{2}{3}$ At Q, $x=0 \therefore y = -2e^3 \therefore Q = (0, -2e^3)$	1 1	Nearly all correct.
d) Area between lines $= \int_0^1 (y_1 - y_2) dx$ $= \int_0^1 e^{3x} - e^3(3x - 2) dx$ $= \left[\frac{e^{3x}}{3} - \frac{3x^2 e^3}{2} + 2e^3 x \right]_0^1$ $= \frac{e^3}{3} - \frac{3e^3}{2} + 2e^3 - \frac{1}{3} = \frac{5e^3}{6} - \frac{1}{3}$ $\therefore$ Area is <u>$\frac{1}{6}(5e^3 - 2)$ sq. units</u>		Poorly done. Diagrams awful. Many people made it difficult by splitting up the area into triangles etc or working off the y axis. Whereas it is a straight forward example of its type.

Suggested Solutions

Marks

Marker's Comments

$$a) \int \frac{x^2+3-8}{x^2+3} dx = \int \left(1 - \frac{8}{x^2+3}\right) dx$$

$$= x - \frac{8}{\sqrt{3}} \tan^{-1}\left(\frac{x}{\sqrt{3}}\right) + c$$

$$b) f(x) = \frac{3x}{x^2+1}$$

$$i) f'(x) = \frac{(x^2+1)3 - 3x(2x)}{(x^2+1)^2} = \frac{3-3x^2}{(x^2+1)^2}$$

$$s.p. \quad -3x^2+3=0 \quad \therefore x^2=1$$

$$\therefore x = \pm 1$$

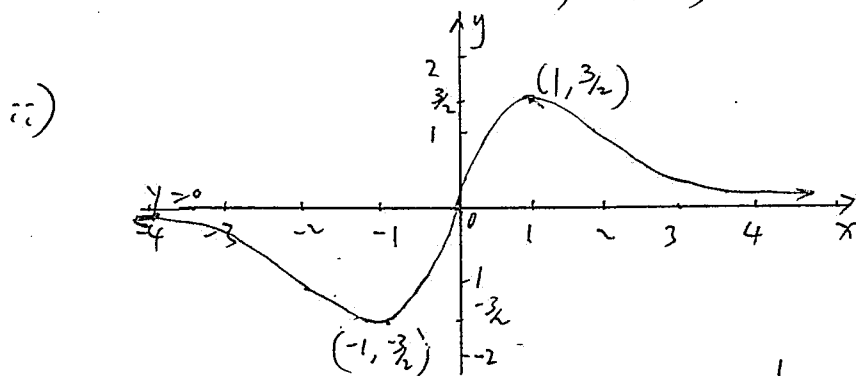
$$\text{when } x=1, \quad y = \frac{3}{2}$$

$$x=-1, \quad y = -\frac{3}{2}$$

Test max/min

x	-2	-1	0	1	2
f'	$-\frac{9}{25}$	0	3	0	$\frac{9}{25}$

rel min at $(-1, -\frac{3}{2})$
rel max at $(1, \frac{3}{2})$



$$iii) \text{Area} = \frac{3}{2} - \int \frac{3x dx}{x^2+1} = \frac{3}{2} - \frac{3}{2} \left[\ln(x^2+1) \right]$$

$$= \frac{3}{2} - \frac{3}{2} \ln 2 \text{ unit}^2$$

#

1 m

1 m

1 m

 $\frac{1}{2}$ m $\frac{1}{2}$ m $\frac{1}{2}$ m $+\frac{1}{2}$ m

2 m

 $1+\frac{1}{2}$ m $\frac{1}{2}$ msame forgot c $-\frac{1}{2}$ m

students have problems with

 $\frac{8}{\sqrt{3}}$ $-\frac{1}{2}$ mjust getting $\int dx = x$
did not score any marks.many forgot y values $-\frac{1}{2}$ mmany students got the wrong values of $f''(1)$ or $f'(1)$, then they wait for the $\frac{1}{2}$ m for each T.P.Forgot write $y=0$ on graph $-\frac{1}{2}$ m.T.P. $\frac{1}{2}$ m $(0,0)$ o shape 1 m
 $y=0$ $\frac{1}{2}$ mForgot unit² $-\frac{1}{2}$ m $\frac{3}{2} \rightarrow \frac{1}{2}$ m $\frac{3}{2} - \int \frac{3x dx}{x^2+1}$ got 1 m

4

MATHEMATICS Extension 1 : Question.....

Suggested Solutions	Marks	Marker's Comments
(a) let $u = 4 - x^2$ $\frac{du}{dx} = -2x$ $du = -2x dx$ $-\frac{1}{2} du = x dx$ when $x=0$, $u=4$ when $x=1$, $u=3$ $\therefore \int_0^1 \frac{x}{\sqrt{4-x^2}} dx = \int_4^3 \frac{-\frac{1}{2} du}{\sqrt{u}}$ $= -\frac{1}{2} \int_4^3 u^{-\frac{1}{2}} du$ $= -\frac{1}{2} \left[2\sqrt{u} \right]_4^3$ $= -(\sqrt{3} - \sqrt{4})$ $= -\sqrt{3} + 2$ $= 2 - \sqrt{3}$ or $= \frac{1}{2} \int_3^4 u^{\frac{1}{2}} du$ $= \frac{1}{2} \left[\frac{2}{3} u^{\frac{3}{2}} \right]_3^4$ $= \sqrt{4} - \sqrt{3}$ $= 2 - \sqrt{3}$	$\left. \begin{array}{l} \frac{1}{2} \\ \frac{1}{2} \\ \frac{1}{2} \\ \frac{1}{2} \end{array} \right\} \frac{1}{2}$	* If you never worked out the new limits - 1mk * If you change the limits but then changed everything back to $x - \frac{1}{2}$ mk
(b) (i) $\sqrt{3} \sin 2\theta - \cos 2\theta = R \sin(2\theta - \alpha)$ $= R \sin 2\theta \cos \alpha - R \cos 2\theta \sin \alpha$ $\therefore \sqrt{3} = R \cos \alpha \quad 1 = R \sin \alpha$ $\tan \alpha = \frac{1}{\sqrt{3}}$ $\alpha = \frac{\pi}{6} \text{ as } 0 \leq \alpha \leq \frac{\pi}{2}$	$\frac{1}{2}$ $\left. \begin{array}{l} \frac{1}{2} \end{array} \right\} \frac{1}{2}$	

2/3

MATHEMATICS Extension 1 : Question 4

Suggested Solutions	Marks	Marker's Comments
$R = \sqrt{3^2 + 1^2}$ $= \sqrt{4}$ $= 2$ $R > 0$ as α is acute, its in first quad	$\frac{1}{2}$	* lost this mark no conclusion
$2 \cdot \sqrt{3} \sin 2\theta - \cos 2\theta = 2 \sin \left(2\theta - \frac{\pi}{6}\right)$	$\frac{1}{2}$	
(10) $\sqrt{3} \sin 2\theta - \cos 2\theta = 1$ $2 \sin \left(2\theta - \frac{\pi}{6}\right) = 1$ $\sin \left(2\theta - \frac{\pi}{6}\right) = \frac{1}{2}$ $2\theta - \frac{\pi}{6} = \frac{\pi}{6} \text{ or } \frac{5\pi}{6}$ $2\theta = \frac{2\pi}{6} \text{ or } \frac{6\pi}{6}$ $\theta = \frac{\pi}{6} \text{ or } \frac{\pi}{2}$ $\left(\frac{\pi}{6}\right) \quad \left(\frac{\pi}{2}\right)$	$\frac{1}{2}$	* If they only said $2\theta - \frac{\pi}{6} = \frac{\pi}{6}$ then; max 1mk * If they used general solutions and made a mistake = 1/2 mks
(c) <u>Method ONE</u> $mx + y = 3 \quad \text{--- (1)}$ $x^2 + y^2 = 5 \quad \text{--- (2)}$ sub (1) into (2) $x^2 + (3 - mx)^2 = 5$ $x^2 + 9 - 6mx + m^2x^2 = 5$ $x^2(1 + m^2) - 6mx + 4 = 0$ line is a tangent to a circle then there's only one soln to quad eqn i.e. $\Delta = 0$ $\therefore (-6m)^2 - 4(1 + m^2)4 = 0$ $36m^2 - 4(4 + 4m^2) = 0$ $36m^2 - 16 - 16m^2 = 0$ $20m^2 = 16$ $m^2 = \frac{16}{20}$	$\frac{1}{2}$	* lost 1/2mk if no explanation!!
$m = \pm \frac{2}{\sqrt{5}}$	$\frac{1}{2}$	* lost 1/2mk if only wrote $m = \frac{2}{\sqrt{5}}$

MATHEMATICS Extension 1 : Question 4.....

Suggested Solutions	Marks	Marker's Comments
(c) <u>METHOD TWO</u> The perpendicular distance from a tangent to a circle equals the radius. $\therefore \sqrt{S} = \frac{ ax+by+c }{\sqrt{a^2+b^2}}$ $\sqrt{S} = \frac{ 0m + 0 \times 1 - 3 }{\sqrt{m^2+1}}$ $\sqrt{S} = \frac{3}{\sqrt{m^2+1}}$ $S = \frac{9}{m^2+1}$ $m^2+1 = \frac{9}{5}$ $m^2 = \frac{4}{5}$ $m = \pm \frac{2}{\sqrt{5}}$ * If the students differentiated = 0 mks.	$\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$	

MATHEMATICS Extension 1: Question.....5

Suggested Solutions

Marks

Marker's Comments

a) $f(x) = \frac{1}{2} \cos^{-1}(3x-1)$

Domain $-1 \leq 3x-1 \leq 1$

$\therefore D = \{x: 0 \leq x \leq \frac{2}{3}\}$

Range $0 \leq \cos^{-1}(3x-1) \leq \pi$

$0 \leq \frac{1}{2} \cos^{-1}(3x-1) \leq \frac{\pi}{2}$

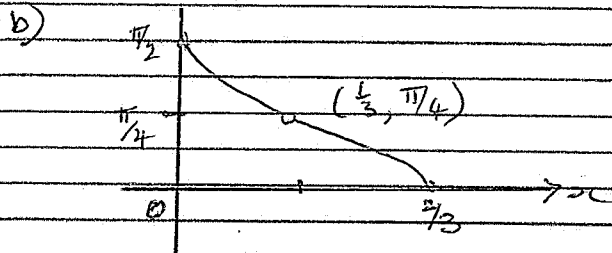
$R = \{y: 0 \leq y \leq \frac{\pi}{2}\}$

①

Domain ①

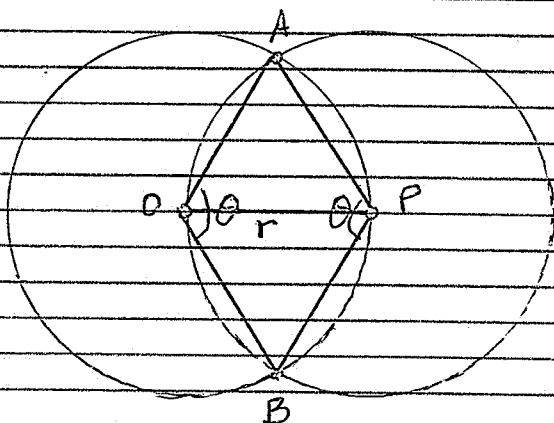
Range ①

correct end points
correct shape
pass through
 $(\frac{1}{3}, \frac{\pi}{4})$
gradient $\neq 0$
at $(\frac{1}{3}, \frac{\pi}{4})$



①

(i)



①

① all sides
equal to r and
circles equal.

① rhombus
reason.

Geometric statements
not in "Geometry notes"
NOT ACCEPTED.

$OA = OB = r$ (equal radii of circle)

$PA = PB = r$ (equal radii of circle)

$\therefore OA = OB = PA = PB$ (equal circles)

$\therefore OAPB$ is a rhombus (4 equal sides)

(ii) Various solutions

$AO = OP = AP = r$

$\therefore \angle AOP = \frac{\pi}{3}$ (angle of equilateral triangle)

$\angle AOB = 2 \times \frac{\pi}{3} = \frac{2\pi}{3} = \angle APB = \theta$

$\therefore \text{Segment } AOB = \text{Segment } APB.$

$= \frac{1}{2} r^2 (\theta - \sin \theta)$

$= \frac{1}{2} r^2 (\frac{2\pi}{3} - \sin \frac{2\pi}{3})$

$= \frac{1}{2} r^2 (\frac{2\pi}{3} - \frac{\sqrt{3}}{2})$

Area
(shaded) $= \pi r^2 - 2 \times \frac{1}{2} r^2 (\frac{2\pi}{3} - \frac{\sqrt{3}}{2})$

$= \pi r^2 - \frac{2\pi r^2}{3} + \frac{\sqrt{3} r^2}{2}$

Area $= \frac{\pi r^2}{3} + \frac{\sqrt{3} r^2}{2} u^2$

① angle with
reason.

① sub into formula.

① $\sin \frac{2\pi}{3} = \frac{\sqrt{3}}{2}$

① correct answer.

MATHEMATICS Extension 1 : Question.....5

Suggested Solutions

Marks

Marker's Comments

$$c) (i) \frac{d}{dx} \left(\tan^{-1} \left(\frac{x}{3} \right) \right)^2 = 2 \tan^{-1} \left(\frac{x}{3} \right) \times \frac{\frac{1}{3}}{1 + \left(\frac{x}{3} \right)^2}$$

$$= \frac{6 \tan^{-1} \left(\frac{x}{3} \right)}{x^2 + 9}$$

(2)

(1/2) each part
of answer.
(1/2) + (1/2)
(1/2) simplified
answer.

$$(ii) \frac{1}{6\pi} \int_0^{\sqrt{3}} \frac{\tan^{-1} \left(\frac{x}{3} \right)}{x^2 + 9} dx$$

(2)

$$= \frac{1}{6\pi} \left[\left(\tan^{-1} \left(\frac{x}{3} \right) \right)^2 \right]_0^{\sqrt{3}}$$

$$= \frac{1}{6\pi} \left[\left(\tan^{-1} \left(\frac{\sqrt{3}}{3} \right) \right)^2 - \left(\tan^{-1} 0 \right)^2 \right]$$

$$= \frac{1}{6\pi} \left[\left(\tan^{-1} \left(\frac{1}{\sqrt{3}} \right) \right)^2 \right]$$

$$= \frac{1}{6\pi} \left[\left(\frac{\pi}{6} \right)^2 \right]$$

$$= \frac{1}{6\pi} \times \frac{\pi^2}{36}$$

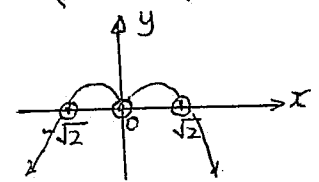
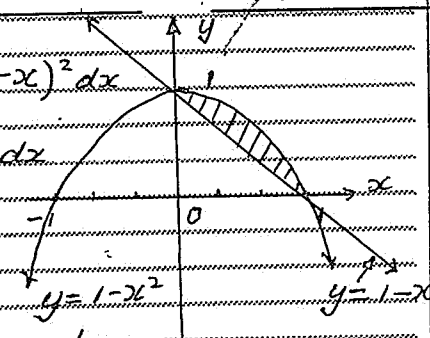
$$= \frac{\pi}{216}$$

$$(1/2) \frac{1}{6\pi}$$

$$(1/2) \left(\tan^{-1} \left(\frac{x}{3} \right) \right)^2$$

$$(1/2) \tan^{-1} \left(\frac{1}{\sqrt{3}} \right) = \frac{\pi}{6}$$

(1/2) correct
answer.

MATHEMATICS Extension 1 : Question 6		
Suggested Solutions	Marks	Marker's Comments
(a) $f(x) = \sin^{-1}(x^2 - 1)$ (i) $f'(x) = \frac{1}{\sqrt{1-(x^2-1)^2}} \times 2x$ $= \frac{2x}{\sqrt{1-(x^4-2x^2+1)}}$ $= \frac{2x}{\sqrt{2x^2-x^4}}$ $= \frac{2x}{ x \sqrt{2-x^2}} = \begin{cases} \frac{2}{\sqrt{2-x^2}}, & x > 0 \\ -\frac{2}{\sqrt{2-x^2}}, & x < 0 \end{cases}$ (ii) Domain: $2-x^2 > 0$ $\therefore -\sqrt{2} < x < \sqrt{2}$ $\therefore \text{Domain} = \{x : -\sqrt{2} < x < \sqrt{2}, x \neq 0\}$	$\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$	Students failed to learn the correct formula for the derivative of $\sin^{-1}f(x)$. 11 Note: $\sqrt{x^2} = x$ Alternatively: $2x^2 - x^4 > 0$ $x^2(2-x^2) > 0$  $\{-\sqrt{2} < x < 0\} \cup \{0 < x < \sqrt{2}\}$ 11
b) $V = \pi \int_{-1}^1 (1-x^2)^2 - (1-x)^2 dx$ $= \pi \int_{-1}^1 x^4 - 3x^2 + 2x dx$  $= \pi \left[\frac{x^5}{5} - \frac{3x^3}{3} + \frac{2x^2}{2} \right]_{-1}^1$ $= \pi \left[\left(\frac{1^5}{5} - 1^3 + 1^2 \right) - \left(\frac{(-1)^5}{5} - (-1)^3 + (-1)^2 \right) \right]$ $\therefore V = \frac{\pi}{5}$ Thus, the volume of the solid of revolution is $\frac{\pi}{5}$ cubic units	1 $\frac{1}{2}$ $\frac{1}{2}$	b) Students failed to learn correct formula and how to apply it correctly for the difference between 2 volumes - Some students misread the question. $\frac{1}{2}$ mark if students vaguely knew that volume had π, a squared term & that the question involved the difference. $\frac{1}{2}$ mark for integrating an expression of equal difficulty.

MATHEMATICS Extension 1 : Question...6

Suggested Solutions	Marks	Marker's Comments
(c) RTP $\tan^{-1} \frac{x}{\sqrt{a^2-x^2}} = \sin^{-1} \frac{x}{a}$ for $x < a$, $a > 0$ APPROACH I: for $-a < x < a \Rightarrow -1 < \frac{x}{a} < 1$ Let $\theta = \sin^{-1} \frac{x}{a}$ $-\frac{\pi}{2} < \theta < \frac{\pi}{2}$ $\Rightarrow \sin \theta = \frac{x}{a}$ For case: $0 \leq x < a$ $0 \leq \theta < \frac{\pi}{2}$ $\therefore \tan \theta = \frac{x}{\sqrt{a^2-x^2}}$ $\therefore \theta = \tan^{-1} \frac{x}{\sqrt{a^2-x^2}}$ For case: $-a < x \leq 0$, $\sin^{-1}$ and $\tan^{-1}$ case $\therefore \tan^{-1} \frac{x}{\sqrt{a^2-x^2}} = \sin^{-1} \frac{x}{a}$ for $x < a$ APPROACH II CONSIDER $y = \tan^{-1} \frac{x}{\sqrt{a^2-x^2}} - \sin^{-1} \frac{x}{a}$ $\therefore \frac{dy}{dx} = \frac{1}{1+\frac{x^2}{a^2-x^2}} \times \frac{a^2-x^2}{a^2-x^2} - \frac{1}{\sqrt{a^2-x^2}}$ $= \frac{(a^2-x^2) \times a^2}{a^2(a^2-x^2)\sqrt{a^2-x^2}} - \frac{1}{\sqrt{a^2-x^2}}$ $= \frac{1}{\sqrt{a^2-x^2}} - \frac{1}{\sqrt{a^2-x^2}} = 0$ $\therefore y$ is a constant for $x < a$ Let $x=0$ $y = 0 - 0 = 0$ $\Rightarrow \tan^{-1} \frac{x}{\sqrt{a^2-x^2}} = \sin^{-1} \frac{x}{a}$ for $x < a$ APPROACH III Let $\alpha = \tan^{-1} \frac{x}{\sqrt{a^2-x^2}}$, $\beta = \sin^{-1} \frac{x}{a}$ consider $\sin(\alpha - \beta)$ etc !!	$\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$	$a^2 = x^2 + \theta Q^2$ (Pyth Thm) $\theta Q = \sqrt{a^2-x^2}$ ADD FNS OF $\sqrt{a^2-x^2}$ θ $x (< 0)$ 2

6
MATHEMATICS Extension 1 : Question.....

Suggested Solutions	Marks	Marker's Comments
(d) Profit $P(x) = R(x) - C(x)$ $= x^2 - 80500 - (87000 + 150x)$ $P(x) = x^2 - 150x - 167500$ Data: $\frac{dx}{dt} = 300$ when $x = 500$ ATF Rate $\left(\frac{dP}{dt}\right)$ when $x = 500$ Now $\frac{dP}{dt} = \frac{dP}{dx} \times \frac{dx}{dt}$ $= (2x - 150) \times \frac{dx}{dt}$ when $x = 500$ $\frac{dP}{dt} = (2 \times 500 - 150) \times 300$ $= 255000$ Rate is \$255 000 / week at 500 units	$\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$	
OR PROFIT $P(x) = R(x) - C(x)$ $= x^2 - 80500 - (87000 + 150x)$ $P(x) = x^2 - 150x - 167500$ Data: $\frac{dx}{dt} = 300$ when $x = 500$ $\therefore x = \int 300 dt = 300t + C$ take $t=0$ $x=500 \Rightarrow C=500$ $\therefore x = 300t + 500$ $\therefore P(t) = (300t + 500)^2 - 150(300t + 500) - 167500$ $= 90000t^2 + 255000t + 7500$ $\frac{dP}{dt} = 2(300t + 500) \times 300 - 150 \times 300$ $= 180000t + 255000$ at $t=0$ $\frac{dP}{dt} = 2 \times 500 \times 300 - 150 \times 300$ $= 255000$ $\therefore$ Rate is (\$255 000 / week at 500 units	180000t + 255000	3

MATHEMATICS EXTENSION 1: Question 7

Suggested Solutions	Marks	Marker's Comments
(a) METHOD 1: SOLVE GRAPHICALLY $x + \frac{1}{ x } < 0$ $x^2 - x - 2 > 0$ $(x-2)(x+1) > 0$ Satisfied $x < -1$ Satisfied $x < -1$ or $x > 2$. $\therefore$ Both inequalities satisfied when $x < -1$.	1 1 1	Solve $x + \frac{1}{ x } < 0$ Solve $x^2 - x - 2 > 0$ (Students who solved algebraically usually mishandled the absolute value.) Combine solutions from both inequalities
METHOD 2: SOLVE ALGEBRAICALLY $x + \frac{1}{ x } < 0 \quad \{x \neq 0\}$ Case 1: $x > 0$ Case 2: $x < 0$ $x + \frac{1}{x} < 0$ $x + \frac{1}{(-x)} < 0$ $x^2 + 1 < 0$ $x - \frac{1}{x} < 0$ $\therefore$ No $\mathbb{R}$ soln. $x^2 - 1 > 0$ * $\therefore x < -1$ or $x > 1$ But $x < 0$, $\therefore x < -1$ only. $x^2 - x - 2 > 0$ $(x-2)(x+1) > 0$ $x < -1$ or $x > 2$ $\therefore$ Both inequalities satisfied when $x < -1$.	1	Students who did not state their cases clearly often confused themselves. * Note change of inequality direction as $x < 0$

MATHEMATICS EXTENSION 1: Question

Suggested Solutions

Marks

Marker's Comments

$$(b) \quad y = \sqrt{x} e^x$$

$$y^2 = x e^{2x}$$

$$\therefore V = \pi \int_1^3 x e^{2x} dx$$

x	1	$1\frac{1}{2}$	2	$2\frac{1}{2}$	3
y^2	e^2	$\frac{3e^3}{2}$	$2e^4$	$\frac{5e^5}{2}$	$3e^6$

$$\therefore h = \frac{1}{2}$$

$$V \approx \pi \times \frac{(\frac{1}{2})}{2} \left[e^2 + 2 \left(\frac{3e^3}{2} + 2e^4 + \frac{5e^5}{2} \right) + 3e^6 \right]$$

$$\approx \frac{\pi e^2}{4} (1 + 3e + 4e^2 + 5e^3 + 3e^4)$$

$$V \approx 1758.0277 \dots$$

Volume is approx 1758 units³
(4 significant figures)

$$\text{NB. } (a^2 + b^2 + c^2 + d^2) \neq (a + b + c + d)^2$$

Furthermore, Volume $\neq \pi \times \text{Area}$

1

Exact expression for volume

1

Application of Trapezoidal Rule
(several used all or part of Simpson's Rule instead)

1

Approximated solution

Many students elected to square results after taking the sum, rather than squaring first

Others calculated the area under the curve and then multiplied by π , forgetting that $V = \pi \int_a^b y^2 dx$

MATHEMATICS EXTENSION 1: Question 7

Suggested Solutions

Marks

Marker's Comments

(c)(i) Desert distance $CD = \sqrt{25+x^2}$

Road distance $BC = 5-x$

$$\therefore T = \frac{\sqrt{25+x^2}}{16} + \frac{5-x}{39} \quad \text{--- ①}$$

$$= \frac{39\sqrt{25+x^2} + 16(5-x)}{16 \times 39}$$

$$= \frac{39\sqrt{25+x^2} - 16x + 80}{624} \quad \text{as required}$$

$\frac{1}{2}$

Desert travel time

$\frac{1}{2}$

Road travel time

(ii) $\frac{dT}{dx} = \frac{x}{16\sqrt{25+x^2}} - \frac{1}{39}$ from ①

S.P exist when $\frac{dT}{dx} = 0$

$$\frac{x}{16\sqrt{25+x^2}} - \frac{1}{39} = 0$$

$$39x = 16\sqrt{25+x^2}$$

$$1521x^2 = 256(25+x^2)$$

$$1265x^2 = 6400$$

$$x^2 = \frac{1280}{253}$$

Since $0 \leq x \leq 5$, $x = \sqrt{\frac{1280}{253}}$

$\frac{1}{2}$

Derivative

$\frac{1}{2}$

Stationary point

x	2	$\sqrt{\frac{1280}{253}}$	3
$\frac{dT}{dx}$	-0.002	0	0.007

$\therefore$ Relative minimum

$\frac{1}{2}$

Test nature of SP

Since this is the only SP & the function is continuous in the domain $0 \leq x \leq 5$, the relative minimum is also the absolute minimum.

$\therefore$ Quickest journey occurs when

$$x = \sqrt{\frac{1280}{253}}, \text{ which results in a}$$

time of 0.413 hours (24.8 minutes).

$\frac{1}{2}$

Resultant time

(Many students wrongly assumed domain as $x > 0$.)